



"Seeing, hearing and feeling are miracles."
Walt Whitman

02 Knowing through our senses.

- 1 Introduction
- 2 Knowledge
- 3 Senses as faculties, sensations as products
- 4 External senses
- 5 Internal senses
- 6 Differences between sensations and perceptions in man and animals
- 7 Characteristics of the sensitive knowledge
- 8 Link between the degrees of life and degrees of knowledge

1 Introduction

In the previous unit we saw that animal life adds to vegetative life local movement. This is possible because animals feel desires and knowledge allows them to find where to quench their desires. In this Page unit we discuss the sensitive knowledge. Animal knowledge, sensitive knowledge is present also in all human beings but is different from animal knowledge because it is guided and used by our superior faculties of knowledge and happens simultaneously. We tend to think animals see the world as we see it, but this is not the case. To transpose our ways of doing things to others is called *anthropomorphism* (from *anthros* = man; *morphe* = shape, form), to give human feelings and behaviours to other realities as it is done in comics and cartoons, so successfully.

2 Knowledge

Many lower species of animals are "deaf" in the sense that they do not have organs developed for detecting sound waves. Among these are annelids, molluscs (squids, octopuses, snails, slugs, clams) and more. The issue of whether a species of lower animal can "hear" is often obscured by the fact that many of them can detect the *vibrations* of sound waves, without actually hearing the sound itself. For example, many insects can detect sound through various organs, not all of which are true "ears." Virtually all animals with ears can become deaf from illness, injury, or medical intervention.

<http://library.gallaudet.edu>

To know is some type of assimilation. Is to change in ourselves (subject) which reflects something objective (distinct, external). Unlike the assimilation or nutrition, knowledge is not physical appropriation. When we eat, we destroy what we eat and make it part of us; we assimilate part and dispose of what we cannot assimilate. In knowledge we do not destroy what we know, we make a kind of internal copy of what we know. Knowing is an activity that makes us aware of other things which are not ourselves. In a sense we become what we know, we adapt ourselves to what we know, to the shape of what we know.

If there is no adaptation, we really do not know. When we see a car, somehow, we get its colour and shape within us, we change to something inside us that has that colour and shape. The same when we smell something, we get the fragrance within us. For us it is difficult to distinguish between what we sense with each sense from what we think or remember, or even what we wish because we do all activities simultaneously and continuously. Only by reflecting and analysing (dividing, separating) and isolating our sensations we are able to distinguish them and compare them with similar sensations in other people and animals. Knowledge in persons can be sensitive, intellectual and personal but for the time being we limit ourselves to the sensitive knowledge.

3 Senses as faculties, sensations as products

We have clear notion of what is to see, and that it is linked to the eyes. This is obvious if we close our eyes we do not see. People without eyes do not see. But there are people who have eyes and do not see. Sometimes it is because they have had a sickness which damages the connecting nerves between the eyes and the brain. At other times it is because a particular area of the brain is damaged. This we do not see but medical science probes it, and sometimes is able to fix the problem and people recover their sight. Nevertheless, what see are not only the eyes, the nerves or the nerve cells in the brain also make it possible for us to see. We have to distinguish between the organs (anatomical and physiological base of our knowledge) and the knowledge itself, which is immaterial. The organs are made for that function, which is seeing.

We say that animals have different knowledge functions or powers which allow them to perform different acts through them. In philosophical terms these powers are called **faculties** and we distinguish them by what they can specifically do, which we call their *specific object*. We say that some animals have the power of seeing and the object of this faculty is what has some *colour*.

The knowledge faculties in animals are called **senses**. The knowledge we get from senses give are called sensations. Sometimes people also use the word “feelings” to mean a particular sensation, though it also may mean a subjective reaction to knowledge, like feeling angry. Because of this we will use the term **sensation** to mean the immediate knowledge given by the external senses.

From Aristotle (Greek philosopher, IV BC) the most common classification is to distinguish between **external senses**, which give us direct impressions of reality and **internal senses**, which elaborate the sensations given by the external senses.

4 External senses

Aristotle speaks of five external senses, three of them simple: touch, taste, smell and two more elaborate: hearing and vision. The elementary ones need to be in contact with what is being sensed, while the superior ones do not, so they can give information from more objects and more precise because they are not as limited by the size of the receptor.

Alexandra Logue was the sort of finicky eater who drives parents to distraction. At 1 year of age, she wouldn't touch anything except bread and milk. Fifteen years later she couldn't stand such teen-age staples as spaghetti or Coke. She still refuses to eat fish.

"I'm what's known as a '**super taster**,'" said Dr Logue, an experimental psychologist at the State University of New York at Stony Brook and author of "The Psychology of Eating and Drinking" (W. H. Freeman, 1986). Dr Logue's genetic ability to taste flavours that most other people can't taste led to her interest in why children and adults have very different responses to certain foods.

"People say very finicky children are doing it for attention," she said, "but they may actually be tasting different things than we do."

—Lawrence Kutner, "Parent and child," *The New York Times*, January 12, 1989.

External senses are organic. Animals have some specific cells to detect some aspect of reality, be it chemical, mechanical, magnetic, electrical, or light based. These specific organic sensors are called receptors. They usually require a minimum of stimulation to be activated and have a maximum beyond which they may be damaged or, at least are blocked. They are called the lower and upper stimulus threshold.

All animals have sense of *touch*. It is the one that they need physically for their physical survival.

They avoid what damages them physically or chemically and move to what pleases them. The sense of taste is probably present in all animals because it is basic for recognising food as good or bad. Some authors said that all animals have some sense of smell because all need oxygen for their vital operations, though the way they smell differs greatly from one to other. It is common knowledge that not all animals have hearing or vision. Unicellular animals, most insects, many fishes do not have hearing organs, but many feel vibrations. Feeling vibrations is not proper hearing and does not need a specific organ because it can be felt by kinetic (movement) organs similar to those inside our body called proprioceptors. Most unicellular animals, worms, many crustaceans do not have eyes, and therefore vision, though they may distinguish light from darkness thanks to some photo sensitive cells.

External senses give us two types of information, one exclusive to its sense, which is called the proper or exclusive sensation, and others common to all senses. The exclusive sensation is what we describe in the next points when discussing each of the five senses.

4.1 Touch

The sense of touch gives different sensations: pressure, pain, temperature, roughness, size and movement. They can be felt differently but we tend to associate them together because they are associated. Whenever we feel something, we get the pressure, smoothness, temperature, size and movement and if it is intense, pain, and pleasurable if it is mild. There are different nervous terminations under the skin for all these sensations. The more abundant are those for feeling pain. The density of terminations depends on the area; they are more concentrated in fingertips, lips, face, neck, tongue, and feet. Women seem to be more sensitive to touch than men. Their skin is softer, more delicate. This may explain why their clothing is thinner and they usually use handbags. This sensitivity is of great use when taking care of babies, who have such a delicate body and skin.

4.2 Taste

Taste is a sensation given by diluted aliments in the mouth. Scientists say that there are five basic sensations: *salty*, *umami*, *sour*, *sweet*, *bitter*. The umami sensation has been found recently as distinct thanks to the studies done in the East, because is linked to a type of dishes that are common in the East and in fish, shellfish, cured meats, mushrooms, vegetables (e.g., ripe tomatoes, Chinese cabbage, spinach, celery, etc.) or green tea.

We have almost 10,000 taste buds inside our mouths; most of them in the tongue and some on the roof of the mouth (palate). Insects seem to have a highly developed sense of taste. They have taste organs on their feet, antennae, and mouthparts. Fish can taste with their fins and tail as well as with their mouth. In humans, girls have more taste buds than boys. The number of taste buds differs, more taste buds give higher sensations and more discrimination. Studies have shown that around 25% of people are said to be non-tasters, 25% supertasters and 50% medium tasters. The more taste buds you have, the more intensely you perceive tastes, especially bitter ones. People who are particularly sensitive are

called supertasters. These numbers can vary depending on sex and ethnicity. Women are more likely to be supertasters and so are people from Asia, Africa and South America. Among women 35% are supertasters while only 15% among men.

Taste is usually felt together with smell. You can feel how different is the taste of food when you have a cold that blocks your nose.

4.3 Smell

Odour	Example
Camphoric	Mothballs
Musky	Perfume/Aftershave
Floral	Roses
Peppermint	Mint Gum
Ethereal	Dry Cleaning Fluid
Pungent	Vinegar
Putrid	Rotten Eggs

We smell substances that come into contact with our nose in very small particles in gaseous form. There is contact between these substances and us and this is why the sense of smell is placed among the simple senses. While insects smell with their antennae, palps, and receptors in other parts of the body, we have the smell sense receptors in special nervous terminations inside the mucosa of our noses. As for the tongue to be active it has to be wet, this then explains why we need handkerchiefs. Humans have about 10 cm² of olfactory mucosa, whereas some dogs have 170 cm². A dog's olfactory epithelium is also considerably more densely innervated, with a hundred times more receptors per square centimetre. Despite this, humans can still distinguish between 4,000 to 10,000 different types of odours. Our sense of smell is greater in childhood and decreases as we grow. It can also be cyclical; women have it heightened during ovulation.

Sickness	Smell characteristics
Anosmia	Lack of ability to smell.
Hyposmia	Decreased ability to smell.
Dysosmia- Phantosmia	Hallucinated smell.
Dysosmia- Parosmia	Things smell differently than they should.
Hyperosmia	An abnormally acute sense of smell.

In mammals there is an area which has special receptors that is connected to a special zone in the brain, which seems to be related to aggressive behaviour and mating behaviour. This system is called the Accessory Olfactory System and the chemicals that stimulate it are called pheromones (*pherein* = to carry, *hormone* = that which sets in motion).

Pheromones are especially important in insect behaviour because they attract potential mates, mark territory, establish trails, coordinate foraging behaviour, signal alarm, warn of danger, or repel predators.

The sense of smell is very linked to feelings because the nerves terminate in the area of the brain that regulates emotions. This is why a particular smell can stimulate our appetite or remind us of our childhood experiences faster than other sensations.

Women are normally more sensitive to smell than men. They detect smaller changes and more varieties. This is probably why perfumes are so important for them, and maybe also why they have more frequent emotional changes.

dB	Sound
20	Whispered voice.
40	Refrigerator humming.
60	Normal conversation.
80	City traffic.
90	Subway trains, lawn mower, motorcycle; prolonged exposure to any noise above 90 decibels can cause gradual hearing loss.
100	Woodworking shop; recommend avoiding more than 15 minutes of unprotected exposure.
110	Chainsaw; regular exposure of more than one-minute risks permanent hearing loss.
120	Snowmobile.
140	Rock concert, firecracker.

4.4 Hearing

The sense of hearing is the first of the two superior senses.

Because it does not need “to touch” the origin of the noise, it can give information of many sounds at the same time and from different distances and directions. The information it gives it is therefore richer than the elementary senses.

The organ of hearing in humans is in the ears, and more specifically in the internal ear. It consists in a tight membrane that vibrates with sound, amplifies the vibrations through very small bones that move a liquid inside a cavity lined with sensors and filled it with a fluid that contains some particles that move those sensors. Those sensors give us the sensation of balance, which tells us our position relatively to the centre of gravity.

The sensation is called sound, and we can distinguish three elements in it, intensity (loudness measured in decibels), pitch (frequency measured in hertz (Hz) per second) and timbre (which is the special combination of different pitches given by the originating vibrations).

We have a threshold for sounds between 20 hz and 20 khz in frequency or pitch. What is below this we call it infrasonic and what is above we call ultrasonic. Bats and dogs can hear ultrasounds; elephants, giraffes and whales use infrasound to communicate. Our best hearing frequency is between 1-3.5 khz.

The sense of hearing is more developed in those animals that can travel long distances like birds or those that live in the savannah, or have more things to communicate, like humans.

Dr Michael Hunter, of Professor Peter Woodruffs group in the Department of Psychiatry and Division of Genomic Medicine at the University of Sheffield, and co-author of the study explains, "Voices allow the brain to determine various factors about a person's appearance, including their sex, size and age. It is much more complex than most people think and is an extremely important tool for determining someone's identity without having to see them.

"The female voice is actually more complex than the male voice, due to differences in the size and shape of the vocal cords and larynx between women and men, and also due to women having greater natural melody in their voices. This causes a more complex range of sound frequencies than in a male voice.

"When a man hears a female voice, the auditory section of his brain is activated, which analyses the different sounds in order to 'read' the voice and determine the auditory face.

"When men hear a male voice the part of the brain that processes the information is towards the back of the brain and is colloquially known as the 'mind's eye'. This is the part of the brain where people compare their experiences to themselves, so the man is comparing his own voice to the new voice to determine gender.

<http://www.sheffield.ac.uk/mediacentre/2005/422.html>

Women have normally a higher pitch than men, which allows them to be heard better. Girls process the audio information in both brains simultaneously, while men use only one. Girls have also more nervous cells in the left side of the brain where speech is processed and a thicker and richer connection between the two halves of the brain. Women use 5 to 7 tones while men usually do not go beyond 3. When highlighting a point women change the pitch, men the volume. Girls normally have a better hearing than boys from birth, and the hearing loss starts usually after menopause, while in men starts at about 30, being severe after 60.

This is normally due to the growing stiffness of the ear drum.

Being exposed to loud or high pitch noises usually produces hearing loss, cardiovascular dysfunctions, stress, annoyance, and aggression.

4.5 Vision

Vision or sight is the most powerful of our senses because it is the one who gives us more information of more reality at the same time. To see how important this sense is one should try do what they normally do with their eyes closed. Though there is a sense compensation on those who do not have sight there is no way they can enjoy the natural sights of the landscape, the colour of flowers and dresses, the paintings and looks of buildings, movies, video games, smiles in people faces, etc.

The organ of seeing is the eyes, which humans have at the front and are placed above the mouth, nose and ears. The eyes occupy one fifth of our face. The eye has about 125 million cells that see in black and white (rods) and with about 7 million that distinguish colours (cones). It allows us to see things in perspective. The region of the brain used for seeing is at the back (occipital) and has more neurons devoted to it than to any other sense. Owls have no cones so they only see in black and white but very well, they can see a mouse moving over 45 feet away with moonlight. Women tend to distinguish colour better than men and have better peripheral and close-range vision, while men seem better at judging movement, focused and long-distance vision.

4.6 Common sensible objects

While each of the senses gives specific information about the world outside us, each gives us some common information. Traditionally they are called the *common sensible* which each gives in a way adapted to its own power. The five common sensible aspects of reality are 1) *size*, 2) *shape*, 3) *number*, 4) *stillness* and 5) *movement/time*.

We have to distinguish well how they show it so as not to confuse what we get from each sense among themselves and also what is proper of the elaboration done by the internal senses on the basic information given by the senses. For example, size of an object seen is extension of colours, as something heard is intensity, as touch is spread, as smell is width with respect to other simultaneous smells, as taste is intensity of flavour. Movement or time for the eyes is changing of colours, in hearing is changing of frequency, for the smell is following the scent, for the taste is the appearance and disappearance of flavour, for touch is the successive change of pressure or heat.

4.7 Objectivity in knowledge

There are many times in which we feel that the eyes give us the wrong information, we have an *illusion*, which is a false vision of reality. There are many types of optical illusions; we can classify them in illusion of: *ambiguity* (you can see two different things), *distortion* (you feel things differently than what they are), *paradox illusions* (you see something that it impossible) and *hallucinations* (when you see what does not exist). Illusions affect other senses but, in less measure, compared to sight, because of the nature of the senses, which are more linked to reality.

Dreams are other experiences in which we “sense” either seeing, smelling, hearing, and tasting things that our senses are not sensing, but we act as if they were there. If we cannot trust our senses in these situations, does it mean we cannot distinguish when we are dreaming from when we are sleeping, and when we are seeing reality or we are seeing an illusion? How can we attain the truth of things? We can distinguish that we are sleeping when we are awake, not when we are sleep. We can see and distinguish illusions from reality by using other senses, but above all by distinguishing what the internal senses tell us, from what the external senses give.

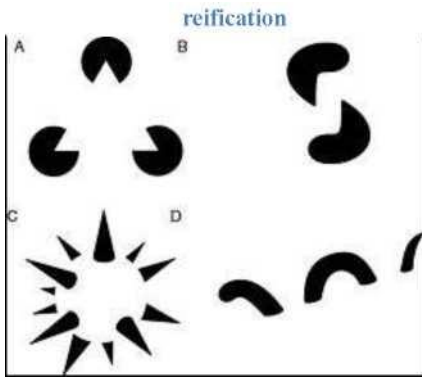
5.0 Internal senses

Traditionally we distinguish four internal senses in progression of depth of knowledge: the *unifying sense*, the *imagination*, *memory* and *cogitative* (estimative in animals).

Emergence



5.1 Unifying sense



Each sense gives us its proper objects; sight gives us colour; hearing sound; smell odour; taste flavour; touch temperature, pressure, and pain. Each cannot tell us that they are activated, because the act of seeing has no colour, the act of hearing makes no sound, and so on. Furthermore, they cannot know what the other senses are sensing; the sight cannot hear, the ear cannot see, the tongue cannot taste, but we sense the colour and crisp sound of the crisps when we are eating and that it belongs to them, not to us. The unifying sense does these two things, senses the senses, and compares and unifies them. By it we have contact with reality. It is the one who is responsible to tell us that we are awake and that the senses are working well and in harmony. The object of the common sense is called **perceptions**, which is how we actually see reality as a unified reality, not as a succession of sensations patched together. Because of this unifying sense we tend to complete objects and associate them according to some rules.

According to this we organise our sensations in the simplest, more regular, way which tends to be symmetric.

“Common Sense is that which judges the things given to it by other senses”, *Leonardo da Vinci*.

The psychological school of perception (*Gestalt* in Germany) analysed some principles which they called *emergence* (the whole is felt at ago, not as a sum of parts), *reification* (we try to make objects out of our sensations), *multi-stability* (Change of perception in ambiguous sensations), *invariance* (we see the same object in spite of rotating, moving or changing size or shades) and *meaning* (we tend to unify sensations to make sense by following the laws of *closeness*, *similarity*, *proximity*, *symmetry*, *continuity* and *common direction*).

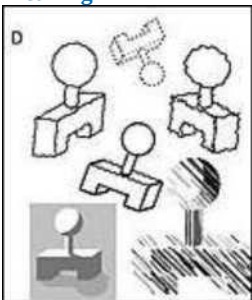
Multi-stability



5.2 Imagination

While all sensations are concrete, specific, singular, and instantaneous to the object sensed, we have other concrete and specific knowledge that occurs when the sensations have disappeared. When we close our eyes we still “see” a kind of picture of what we saw. It is not the same, because we do not keep all what we saw, only the main characteristics, but they are there and are real. Something of what we have seen, heard, smell, taste and touch remains within us in some shape different from the actual sensation. These images are the object of the internal sense called imagination of phantasy (from Greek *phantasm* = image). The images of taste are important for cooking, the images of sound for music, the images of touch for physiotherapists, the images of smell for tea tasters; they refer to them when looking for products that have similar characteristics.

Meaning



The images are not as rich as the sensations, very few people have colour imaginations, or multi-channel sound images (in fact most nightmares are with masculine voices because they are much simpler than women voices). What we keep as images are the perceptions and we further work by combining them following similar rules than those of the perception. The imagination works on perceptions and the images it has kept before.

“Dreams are rough copies of the waking soul”, *Calderon de la Barca*

Depending on this we can distinguish some types of imagination: *pictorial*, *schematic*, *proportional*, *symbolic*, *productive*. *Pictorial* is the closest to the perception or sensations we receive, it is richer in detail and freshness, but has little elaboration, it is very strong in children and is the one used mainly in oral narratives; *schematic*, gives us the outlines or sketches, simplifying details and keeping the most salient features of our perception as in cartoons or drafts; *proportional*, helps us to compare different images of similar perceptions and rank them; *symbolic*, is more elaborated relates several different images to a different image as it happens with the traffic signals and finally *productive or creative imagination*; multiplies and creates new images by combining all previous types.

“The senses do not deceive us, but the judgment does”, *Goethe*

The imagination is a powerful tool, but it can create a world on its own if it is not linked to the real world through the unifying sense that gives us the direct view of reality and is directed by the intelligence to its proper function. Idleness tends to foster the unlinked use of imagination because the senses are not keeping it on the alert, and the intelligence is not giving it a direction. Other ways of delinking the imagination from the senses and the intelligence are drugs, some mental sickness which cause hallucinations, and hypnosis which is a directed state of hallucination.

What we have felt strongly remains in the imagination for a long time, in many cases, for the entire life. This is why it is important to control our experiences, look for what will benefit us for life, not what will pull us down and create conflicts in us.

5.3 Memory

“Own only what you can carry with you; know language, know countries, know people. Let your memory be your travel bag”, *Alexander Solzhenitsyn*.

Higher than the imagination is the memory. It is a different faculty which only adds to the imagination the distinction of timing. It tells us whether the image we are contemplating at a particular moment is more or less actual or we got it before. It can tell us this in a simple way as *recognition* (knowing we had it before), *timing* (when it exactly happened), *reconstruction* (how it happened and associated experiences) and finally if we have trained it properly *recalling* (we can look for and find the images of the type and time we want).

Different people have different types of memory depending on their skills and training; painters and photographers tend to have good visual memory, musicians and actors good hearing memory, cooks and tasters tend to have good smell and taste memory, while gymnastics, tailors, physiotherapists have good touch memory.

Memory is like a good, organised database. What memory adds to the data (the images) is their interconnection with the unifying sense, this is their connection with our evolving chain of sensations in time.

“The sense of smell can be extraordinarily evocative, bringing back pictures as sharp as photographs of scenes that had left the conscious mind”, *Thalassa Cruso*,

The process of improving memory takes into account the three phases of memory, the first is the production of the images in time (*encoding-learning*) the second the positioning of images in the database (*recording memorising*) and if these have been done well the last process of remembering (*recalling*) will be successful. Studies show that the more intense the original sensation is the more one remembers, extreme experiences in life are hardly forgotten for example. Because of this it is recommended to have a very positive attitude, total intensity, and attention to what we want to remember. Regarding recording the methods devised to help us remember it is recommended associating what we are learning to a previous well learned topic, to have a place where to pin our new experiences. Visual people tend to associate new experiences to the places they happened, the city, the house, the room; hearing people to the song, melody, when the event happened, etc. This is also true at the intellectual level, because understanding is to make sense of what new we learn as regards to all our previous knowledge, i.e., to position the new within the old, or to reposition the old with regards to the new truth discovered. Finally recalling has a life on its own.

Researchers discover that we have two types of memories, the short-lived ones, which tend to last a day, and the long-time memory, which lasts much longer. They found that frequent recall, repetition before going to sleep is one of the best practices.

The memory in human beings is different than the memory in animals.

5.4 Estimative

“Common sense is seeing things as they are, and doing things as they should be done”, *Unknown*.

The purpose of sensitive knowledge is the promotion of sensitive life. The function of knowledge is to tell the animal whether what is around him is beneficial or detrimental to his life. None of the senses we have discovered does this, but it is a fact that normally animals know what is good for them and what is bad and move towards or away from them. The estimative is the ingrained faculty that tells the animals what is good or bad for them. This faculty is the most needed. Animals may have some defective external senses, there are deaf dogs, blind cats, etc but they can survive. Without a proper estimative they will not last long. Psychologists speak of instinctive knowledge when they refer to the actions of animals that are inherited, not learned, automatic and happen always and in the same way in the species. The estimative is the origin of these behaviours.

Animals learn, because they have images and they have memories, and it is part of their ingrained behaviour to blend this knowledge in the degree that their estimative allows them depending on the species.

In human beings the instinctive behaviour is superseded very early by the intellectually learned behaviour and this is why

classic authors call this internal sense *cogitative* when they refer to human beings.

6.0 Differences between sensations and perceptions in man and animals

Human beings have senses, both internal and external. None of their senses is top in the animal kingdom. The birds in general have more powerful eyes both for long and short distances and for colours (pigeons have five types of cones to distinguish colours, while we have only three). Bats can hear frequencies between 3 Khz and 120 Khz while we are limited to 16 hz to 16 Khz. Dogs, rats and cows smell thousands of times better than we do. The sense of taste is more developed in cows that have about 25,000 taste buds, rabbits 17,000, while men usually have about 10,000. Finally in touch star-nosed moles have 100,000 nerve fibres. This is almost six times more than the touch receptors in the human hand.

Regarding the internal senses our babies are worse prepared than any animal to survive because their unifying sense, imagination and cogitative are less developed at birth. It takes them longer to stay on their feet, feed by themselves, discover what is good and bad for them as developmental psychology shows.

Nevertheless, no animal can see as far as we can using telescopes and microscopes, no one can hear the sounds of the planets or communicate with cellular phones, no one makes and uses perfumes, no one has degrees in cooking and artificial foods, no one has developed thin and warm clothes and shoes, no one draws cartoons, painting, history and no one has medicine or chemistry.

We use our senses, both internal and external, we know how they work and can improve on them, because we are more than senses. Our sight allows us to relate to others and because of the need we have to communicate we have more facial muscles and use them better than other animals, because our face is the reflection of our soul, we use our hearing for communication, to make others be intimate with us, through language, poetry, music; we use our sense of smell and taste to make others enjoy at the same time as they feed and relate giving us the personal characteristics of each individual, we make an art of what we see, hear, smell, taste or touch. We can do this because our imagination is more than pictorial, schematic, or proportional, it is mainly symbolic and productive; it is creative. Our estimative is more than instinctive, we do not adapt ourselves to the environment we adapt the environment to our needs. This is only possible if we are more than animals. What makes us more than animals is what we shall discuss in the remaining units of this subject.

7.0 Characteristics of the sensitive knowledge

“But where the **senses** fail us, reason must step in”, *Galileo Galilei*

Most manuals say that knowledge, especially sensitive knowledge is passive. Our knowledge reflects what reality is. If there is no colour or light outside us there is no possibility of seeing. We only see when there is enough light and some colour. We also reflect only what is outside, we do not put anything new to the knowledge, and otherwise we will deform the reality and compromise truth. Nevertheless, dead animals do not sense, and each animal senses according to the characteristics of its own senses. Pigeons see the same things differently than how we see them. Also, we see things differently when we have a cold than when we are healthy. We also see things better when we are interested in them when they are vital for us, and when we are focused on a particular one, we see it better and tend not to see other things. There is a great activity in knowledge and the more receptive we are the more we receive; the activity to keep us receptive is a difficult one and this is why animals need to disconnect and sleep, lions sleep over thirteen hours a day, cats twelve, dogs ten hours a day, even insects sleep cockroaches sleep 14 hour each day.

Humans are not determined too much, so they are more plastic, open to many ways of doing things. This is why they can develop their senses in different ways and look for different ways to do other things. This plasticity requires training, education. If the babies are not trained, they will not develop their sensorial powers and they will be in a bad position to confront the world ahead of them. The training for them is basic and starts even before they are born because, as in other mammals, their nervous system is active before they are born. They can feel touch sensation, taste, hear and feel the light. It is not clear whether preborn babies can smell because their sensorial terminations are still embedded in the amniotic liquid.

8.0 Link between the degrees of life and degrees of knowledge

Life is activity, not passivity. Life has some characteristics common to all living things. The more perfect these characteristics are the more “living” is the living being, the more active it is, the more perfect it is.

- 1) *Self-motion*: to move oneself, independence, or autonomy in movement. This is the definition given by Aristotle. Something is alive if the source of its motion is within itself; not moved by another. He referred to life as *first act*; vital operations were called *second acts*, because before one could act, one had to be alive first. Operations are *accidental predicates* of a live being, i.e., the being is alive regardless of whether or not operations are performed, e.g., to nourish oneself, to sense, to move from one place to another, to understand. “To be alive” is, on the other hand, an *essential predicate*: a living being exists if it is alive; if it has no life, it ceases to exist. Plants have some type of self-motion, but it is only for growing not local motion, and they therefore have no knowledge. Human knowledge, even at the sentient level is higher, more active than the knowledge of animals.
- 2) *Immanence*: “to remain in” (in-manere). A living being is not only the principle of his operations, but also controls them, i.e., he benefits from his operations. Immanent actions are those whose effects remain within the agent, e.g., eating, reading, crying, sleeping. The actions of inanimate beings are transient; their existence is spent as time goes

on. Not so with living beings: their actions remain within them, e.g., the memory retains the past. There are different forms of immanence: nourishment remains in the being as part of its physical substance; actions performed remain as habits; knowledge remains as the possession of a form. The richer the knowledge is the more it remains within. It is clear that the imagination and memory of human beings is higher than those of animals, so, they have more internal growth, they are more immanent.

- 3) *Unity*: living beings are one—individual; not just possession of some quantity, like rocks, for example. “To live” and “to be” signifies to be *one*, i.e., this particular living being and not another. The higher the form of life, the more unity it possesses. When the unity is destroyed (for example, when you dismember a man), he also ceases to live. Whereas if you divide a plant, it becomes two different individuals. The unity of knowing oneself is higher than those who do not know themselves. In men this unity is higher, thanks to higher imagination and memory, but it is much higher because of the intelligence and his personal knowledge as we shall see in future units. “Love the animals, love the plants, love everything. If you love everything, you will perceive the divine mystery in things. Once you perceive it, you will begin to comprehend it better every day. And you will come at last to love the whole world with an all-embracing love”, *Fyodor Dostoyevsky*
- 4) *Self-development or self-realization*. Living beings tend towards the perfection or growth proper to their nature. Life is potency in its double meaning of possibility and capacity. To affirm that *I am what I am* (i.e., I am the nature I am supposed to be), then I have to be everything I can be from the very beginning. Since I am not, then self-development constitutes a program of life: I have to be what I am. The self-development is based on self-knowledge. This happens instinctively in animals, but only properly in human beings, who are the only ones who know what they are properly and can chart their future according to their intimate knowledge of where they are made for.
- 5) *Cyclic rhythm*: life manifests itself in repeated cycles—birth, growth, reproduction, death. In human beings this cycle is known, but superseded with spiritual life that has no ends. This is why human beings are the only ones who buried their dead, and worship.

Texts to read:

MIMBI, P., *The Overlooked Factor*; pp. 66-72; 96-100; 102-107; WIKIPEDIA; voices “Sense”; “Perception”, “Eye”, “Taste”, “Smell”